Installation Instruction

Installation Instruction Number: 4908439	Released Date: 27-apr-2015
Lubricating Oil Filter Bypass Valve	

Lubricating Oil Filter Bypass Valve

A new lubricating oil filter bypass valve, Part Number 4357177, has been released.

Product Affected

- All ISX/QSX 11.9 liter engines
- All ISX/QSX 12 liter engines
- All ISX/QSX 15 liter engines

Lubricating oil filter bypass valve, Part Number 4357177, replaced the lubricating oil filter bypass valve, Part Number 3820320.

The new lubricating oil filter bypass valve is backwards compatible with all prior lubricating oil filter cooler housing designs.

Preparatory Steps

Park vehicle on a level surface and in an area where it is safe to idle for an extended period of time.

Turn engine OFF and set vehicle brake(s).

Chock vehicle wheels or use an appropriate anti-roll device to stabilize vehicle.

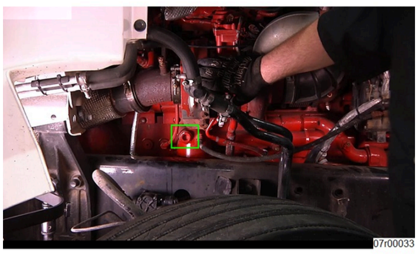
Open vehicle hood.

Disconnect the batteries. See Original Equipment Manufacturer (OEM) service information.

Locating Component

⚠ WARNING ⚠

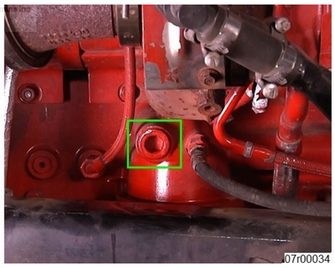
The exhaust and exhaust components can remain hot after the engine has been shut down or secured. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this



procedure or repair and make sure that no combustible materials are located where they might come in contact with hot exhaust or exhaust components

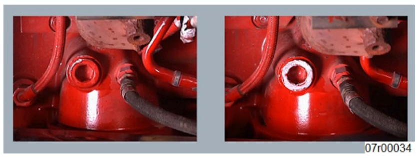
Locate turbocharger on exhaust side of engine.

Locate lubricating oil filter bypass valve retaining plug just below turbocharger.



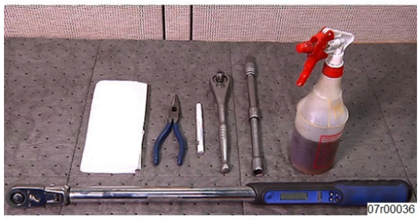
Inspect face of retaining plug for a highly visible paint marking (such as white or yellow paint).

A paint marking indicates the new lubricating oil filter bypass valve has already been installed. No further action is necessary.



Locate the following items in order to complete the repair:

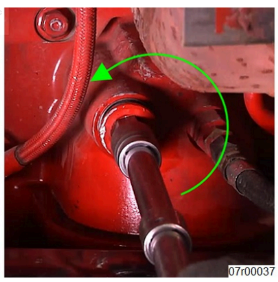
1. Gloves
2. Safety glasses
3. Lint free cloth
4. Paint marker (white or yellow)
5. Needle nose pliers
6. Ratchet wrench and extension
7. 17 mm Allen® wrench or hex socket
8. Torque wrench
9. Clean engine oil
10. Drip pan or absorbent mat



Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



⚠ WARNING ⚠

To reduce the possibility of personal injury, wear protective gloves when handling parts with sharp edges.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

To reduce the possibility of personal injury, wear the appropriate eye and face protection when removing the straight threaded o-ring plug holding in the lubricating oil filter bypass valve spring. The spring is under pressure and can cause personal injury.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

Place an appropriate drip pan or absorbent mat directly under the oil filter in order to properly contain any lubricating oil that may escape during the repair process.

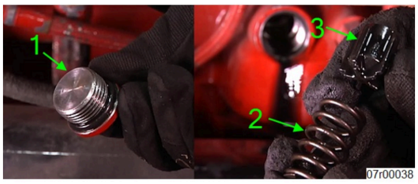
Clean area around the lubricating oil filter bypass valve plug prior to removal.

Use 17mm Allen® wrench or hex socket to loosen the lubricating oil filter bypass valve plug.

Remove retaining plug (1), spring (2), and plunger (3) from lubricating oil cooler housing.

Spring (2) and plunger (3) **must** be discarded.

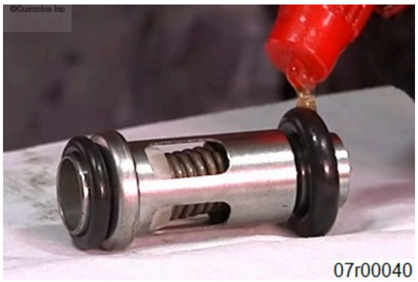
Using a clean lint free cloth, clean retaining plug.



Install

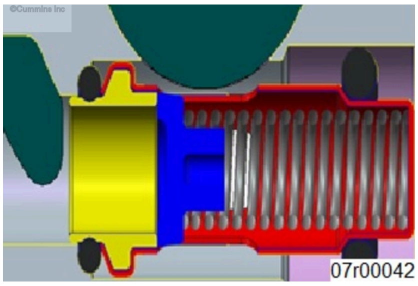
Inspect new lubricating oil filter bypass valve to verify two o-rings are correctly installed as shown.

Lubricate the lubricating oil filter bypass valve with clean engine oil.



Insert lubricating oil filter bypass valve into lubricating oil cooler housing. Marking stating “THIS END OUT” **must** be visible.

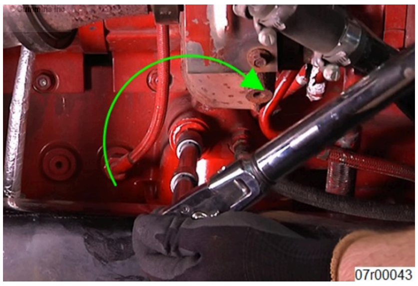
Fully seat lubricating oil filter bypass valve in lubricating oil cooler housing. Verify o-rings are in proper location.



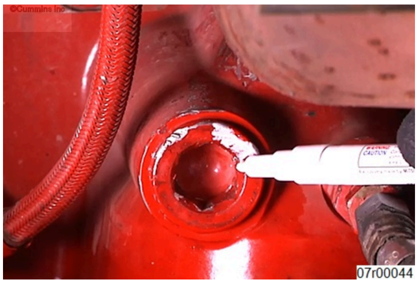
Install retaining plug into lubricating oil cooler housing.

Tighten retaining plug.

Torque Value: 45 n•m [[33] ft-lb]



If installing new lubricating oil filter bypass valve for the first time, clearly mark entire face of retaining plug using a high quality paint marker that is highly visible (such as yellow or white). Liberally apply paint marking.



Finishing Steps

Connect the batteries. See OEM service information.

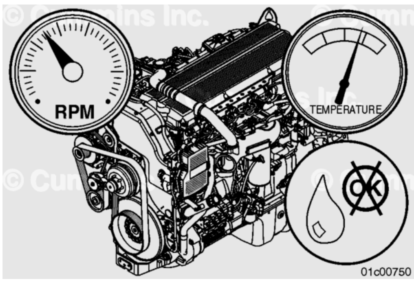
Verify the vehicle brake(s) are set.

Start engine and operate to normal operating temperature. Check for leaks.

If leaks are found, turn engine OFF and verify proper torque procedure was followed.

If leak persists remove plug and inspect plug o-ring and plug for damage.

If no problem is found and oil leak persists please contact a Cummins® certified repair location.



Turn engine OFF and set vehicle brake(s).

Close vehicle hood.

Remove wheel chocks or other anti-roll device.

Document History

Date	Details
2015-4-27	Module Created